

Prompt Engineering

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Week 7: Mid-Semester Review

Course Roadmap

- Weeks 1–3: Foundations, model mechanics, core prompting principles.
- Weeks 4–6: Patterns, frameworks, tone control, iterative refinement.
- Week 7: Consolidate knowledge, surface gaps, plan for project phase.

Session Goals

- Reflect on key insights and lingering questions.
- Practice integrated prompting challenges across domains.
- Assess individual and team readiness for the second half.
- Gather feedback to adjust pacing and support.

Core Concepts Recap

- Prompt anatomy: role, context, constraints, format, tone.
- Model behavior: tokens, attention, capabilities, limitations.
- Pattern stacks and frameworks for repeatable success.
- Iteration loops: diagnose, critique, refine, document.

Common Pitfalls Revisited

- Overloaded prompts without clear priorities.
- Missing guardrails leading to unsafe or off-brand outputs.
- Neglecting evaluation metrics and feedback loops.
- Failing to adjust tone/register for target audiences.

- Rate confidence (1–5) on fundamentals, frameworks, tone control, iteration.
- Identify two strengths and two growth areas.
- Note resources or support needed (tools, exemplars, peer review).
- Share with partner; capture differences in perspective.

Prompting Diagnostic Quiz

- Short answer: define tokenization, pattern stacking, critique prompt.
- Scenario analysis: identify missing components in sample prompts.
- Quick build: draft prompt for given task in 5 minutes; peer review.
- Discuss answers as a class; clarify misconceptions.

Challenge 1: Cross-Domain Brief

- Teams receive multi-part scenario (e.g., healthcare + policy + comms).
- Deliverables: longform summary, patient-facing FAQ, executive brief.
- Must reuse prior frameworks, tone guides, and iteration loops.
- Submit outputs, critique notes, and final revisions.

Challenge 2: Debugging Lab

- Provided with flawed AI outputs and prompt history.
- Diagnose issues, propose prompt edits, run improved version.
- Document fix rationale and residual risks.
- Share quick postmortem with class.

Course Feedback Pulse

- What topics need deeper coverage or more practice?
- Which tools or readings have been most/least helpful?
- How is workload and pacing? Adjustments needed?
- Collect anonymous feedback via quick survey.

Looking Ahead

- Week 8: Evaluating AI outputs with qualitative and quantitative methods.
- Week 9: Multimodal prompting (text + image + audio inputs).
- Week 10+: Applied problem solving, domain projects, ethics.
- Midterm project kickoff next week.

Assignment: Mid-Semester Reflection Dossier

- Compile a 4-page dossier containing:
 - Summary of key takeaways from Weeks 1–6.
 - Two prompts demonstrating progress (before/after).
 - Personal learning goals for Weeks 8–14.
 - Feedback for instructor (what to start, stop, continue).
- Optional: include prompt metrics or workflow diagrams.

- **Completeness:** All sections addressed with thoughtful detail.
- **Insight:** Demonstrates reflection on strengths and gaps.
- **Evidence:** Includes concrete examples or artifacts.
- **Actionability:** Sets specific goals for upcoming modules.

Key Takeaways

- Reflection consolidates knowledge and reveals blind spots.
- Integrated challenges mirror real-world, multi-constraint tasks.
- Feedback loop between students and instructor keeps course agile.
- Prepare for deeper evaluation, multimodal, and ethics modules.

Thank You!

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Questions?